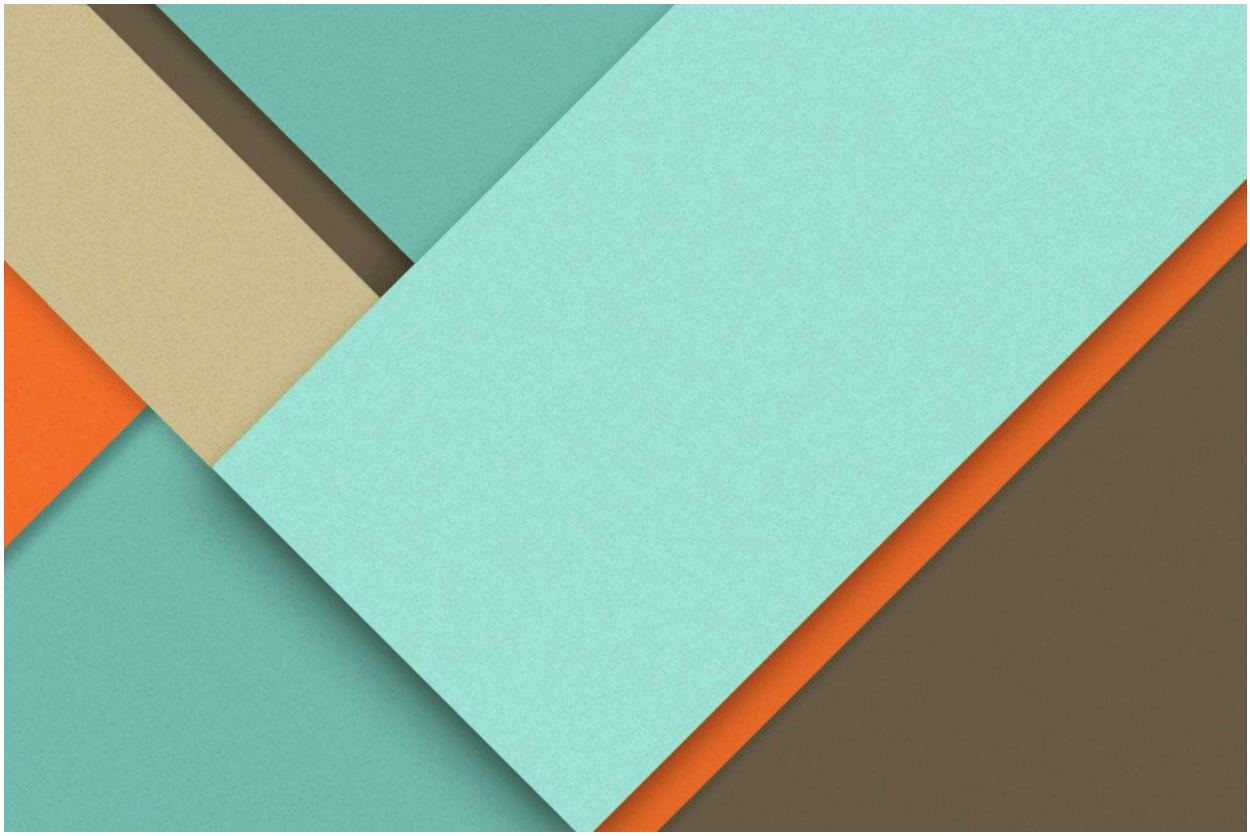


Weber State Registration System Card Sorting Experiment

11/05/2026

Logan Smith
https://github.com/ljswebdev/logan-smith-hci-phase-2

Background and Purpose

This experiment was conducted in order to gain empirical user data regarding Weber State's Class Registration System. I've tasked four participants of varying ages and educational backgrounds to sort and categorize 20 distinct cards containing features and tasks offered by the Class Registration System as a whole. It is my intention to draw insight by comparing and contrasting each of their unique data sets.

Each of the sorting activities was done on a Monday.com Kanban board, beginning with all 20 tasks belonging to a single initial category titled "Tasks"

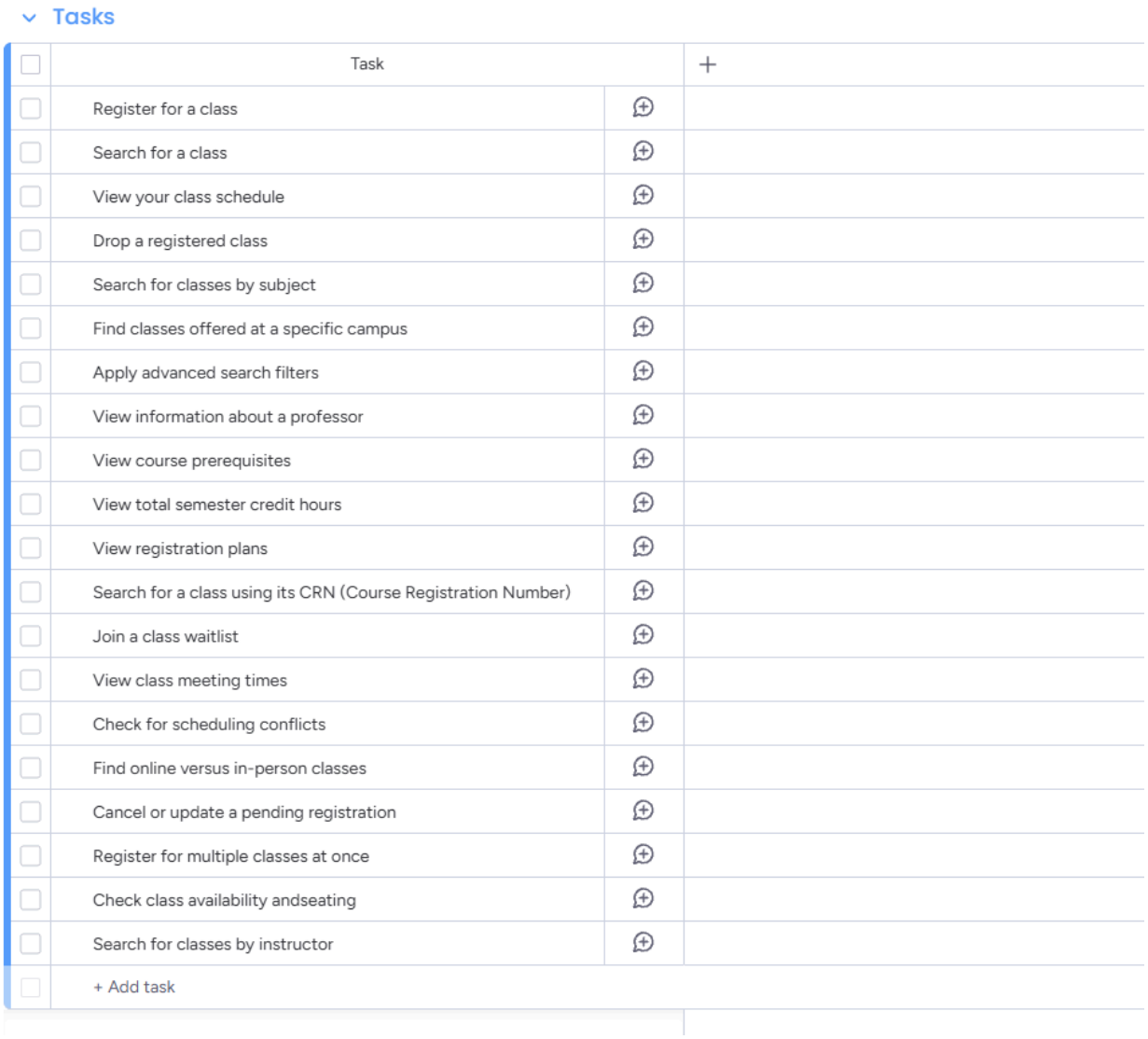
Instructions

After familiarizing each participant with the Kanban board and ensuring they understood how to create groups and sort tasks, these are the exact instructions I provided:

- You will be given 20 distinct task cards representing features and functionalities that Weber State's Class Registration System provides.
- I want you to organize each of these cards into groups based on how you believe they belong together.
- After you have assigned the groups, please label each group with a name and / or description.
- Again, please organize the cards based on how you would expect to find these features and tasks when using the Class Registration System.
- There are no right or wrong answers. Have fun with it!

Starting Point

This screenshot is the initial starting point for each of the participants - 20 cards belonging all belonging to the group “Tasks”



Participants

Carissa, 31 year old female

Carissa is a registered Weber State student who has used the registration portal before, but has taken two semesters off and is planning on returning to Weber State for the Spring 2027 semester.

Crystal, 66 year old female

Crystal is not a college student but will be representative of a prospective new student who has never used the Class Registration System before.

Amy, 31 year old female

Amy attended Weber State over a decade ago and will represent a student who used the Class Registration System in the past.

Hank, 33 year old male

Hank graduated from Utah State University and will represent a student who has used a different university's class registration system.

Carissa's Card Sorting Study

My schedule

Task
View your class schedule
View total semester credit hours
Check for scheduling conflicts
Find online versus in-person classes
+ Add task

Finding my classes

Task
Search for a class
Search for classes by subject
Find classes offered at a specific campus
Apply advanced search filters
Search for a class using its CRN (Course Registration Number)
Search for classes by instructor
+ Add task

Planning my return

Task
View registration plans
Register for multiple classes at once
+ Add task

Course info

	Task		
	View information about a professor		
	View course prerequisites		
	View class meeting times		
	Check class availability and seating		
	+ Add task		

Managing registration

	Task		
	Register for a class		
	Join a class waitlist		
	Cancel or update a pending registration		
	Drop a registered class		
	+ Add task		

Crystal's Card Sorting Study

Finding the right class

Task
Search for a class
Find classes offered at a specific campus
View information about a professor
View course prerequisites
View class meeting times
Search for classes by instructor
Apply advanced search filters
Search for classes by subject
+ Add task

Signing up for classes

Task
Register for a class
Search for a class using its CRN (Course Registration Number)
Join a class waitlist
Cancel or update a pending registration
Drop a registered class
+ Add task

Viewing classes

Task
View your class schedule
Check for scheduling conflicts
Find online versus in-person classes
Check class availability and seating
View total semester credit hours
+ Add task

Planning

Task
View registration plans
Register for multiple classes at once
+ Add task

Amy's Card Sorting Study

Catalog

Task	
Search for a class	
Find classes offered at a specific campus	
View information about a professor	
Search for classes by instructor	
Search for a class using its CRN (Course Registration Number)	
Search for classes by subject	
View course prerequisites	
+ Add task	

Registration

Task	
Register for a class	
Join a class waitlist	
Cancel or update a pending registration	
Drop a registered class	
+ Add task	

Scheduling

	Task	
☐	View your class schedule	
☐	View class meeting times	
☐	Register for multiple classes at once	
☐	Find online versus in-person classes	
☐	Check for scheduling conflicts	
☐	+ Add task	

Student Information

	Task	
☐	View total semester credit hours	
☐	View registration plans	
☐	+ Add task	

Class Search

	Task	
☐	Check class availability and seating	
☐	Apply advanced search filters	
☐	+ Add task	

Hank's Card Sorting Study

Course Search

Task
Search for a class
Search for classes by subject
Find classes offered at a specific campus
Apply advanced search filters
Search for a class using its CRN (Course Registration Number)
Check class availability and seating
Search for classes by instructor
+ Add task

Course Details

Task
View information about a professor
View course prerequisites
View class meeting times
+ Add task

Enrollment Management

Task	
Register for a class	
Drop a registered class	
Join a class waitlist	
Register for multiple classes at once	
Cancel or update a pending registration	
+ Add task	

My Schedule

Task	
View your class schedule	
Check for scheduling conflicts	
Find online versus in-person classes	
+ Add task	

Semester Planning

Task	
View registration plans	
View total semester credit hours	
+ Add task	

Hank's Card Sorting Study

Participant Groupings and Common Patterns

I noticed several similarities in how each of the participants organized their selection of 20 cards, but one of the most consistent patterns was that all four participants categorized cards into a group related to finding or searching for classes. For example, Carissa, Amy and Hank all grouped searching for classes by subject, campus, instructor and CRN together, and Crystal had similar categorization results. This finding suggests that most users prioritize search tools or features that assist in finding courses to be the most recognizable. I also noticed that class registration was another common theme among the sorted card results.

There was also strong similarity between Carissa's and Amy's results, especially regarding scheduling functionality. Both participants created a separate category for managing their class schedules ("My Academic Schedule" and "Class Scheduling", respectively) They both grouped viewing a class schedule, checking schedule conflicts and identifying online vs. in-person classes together. Hank and Crystal had similar but slightly different approaches in their "My Schedule" and "Managing my classes" categories. I took away from these findings that the two most important features are searching for classes and registering for classes.

Unexpected User Mental Models

One thing I found interesting was how much a participant's previous experience influenced their groupings, and I saw a positive correlation between experience with Weber State's Class Registration System and accuracy in categorizing the tasks to the real-life application's model. Carissa tended to organize her cards to suit her mindset as an experienced, returning student, and even created the category "Planning My Return" This was quite different when compared to Amy's groupings, as Amy distinctively separated academic information and available classes into their own categories. This suggests that even if users share similar college experience backgrounds, they can group tasks very differently based on where they currently stand in their academic journey.

Hank's groupings were similar to Carissa and Amy's, but he approached the sorting activity by organizing around individual functions. Seeing that Hank has already graduated from a different university and has vast experience with a completely different class registration system, it would make sense that his choices would be structured differently.

Finally, Crystal's results were noticeably different. She created four broad categories ("Finding the right class", "Signing up for classes", "Managing my classes" and "Planning my education") and the lack of specification seemed to fall in line with someone unfamiliar with the registration process. Crystal appeared to sort and categorize her cards based on her own desires and curiosity, rather than splitting them into individual system functions like the other participants tended to do. This highlights my above findings that the more college education and/or experience with a registration system a user has, the narrower and more defined their card sorting tends to be.

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Similarity Matrix

Feature or Grouping	**Results**	**Level of Agreement**
Finding courses	All four participants created a category that covered finding or searching for a class	Strong
Registration	All four participants grouped registering, dropping, waitlisting and cancelling / updating a registration together	Strong
Schedule	All four grouped schedule viewing, conflict resolution and online vs. in-person class scheduling together	Strong
CRN Search	Carissa, Amy and Hank grouped CRN search together, but Crystal	Some

	grouped it with signing up for classes.	
Class availability	Carissa grouped class availability task cards with course information, Crystal grouped it with class management, Amy grouped it with finding available classes and Hank grouped it with course search.	Strong
Course details	All four grouped professor information and course prerequisites together.	Strong
Registration planning	All four participants grouped registration plans with either academic planning or information.	Middle-ground

Post-Mortem Reflections

How did the empirical data from your card sorting experiment challenge your initial assumptions about how the system's navigation should be structured?

My initial assumption was that all four participants' card sorting results would be very similar, especially Carissa, Amy and Hank's. I expected Crystal's results to be the most unique, which they largely were, but this experiment gave me a lot more insight into **why** users think a certain way. I had failed to realize that even if users have similar experiences and backgrounds, and even if users are familiar with the same systems, they will still approach the same task with a completely different mindset and produce widely varying results.

While generally expected and understood concepts such as searching and registering for a class were frequently split off into their own categories, more specific features were commonly recategorized into other groupings. For example, class seat availability features were grouped differently amongst the participants (Hank grouped it under Course Search while Crystal grouped it under Viewing Classes). Class seat availability is a vital part of the registration process as it can make or break if a registration ever happens. But no matter how important a feature is, users may expect it to belong to a completely different workflow / UI grouping. It really makes me think about how designers decide on where to put features based on their empirical research findings. Perhaps some companies tally up the total number of aligned card groupings and go with the highest number, but what do they do if it's a 50/50 split?

Describe how your annotated ARIA landmarks and keyboard tab sequences ensure that screen reader users navigate your revised information architecture as efficiently as sighted users.

The annotated ARIA landmarks enable sight-impared users to understand the structure of the HTML page without ever having to see it thanks to the innovation of screen reader technology. Using ARIA tags, labels and descriptions, screen readers are able to distinctively identify and explain the most important sections of a page without ever having to see it. I experimented with this myself by closing my eyes and relying on my computer's screen reader and the keyboard, and while it was initially difficult, I was amazed at how accurate and helpful the technology was in navigating the web page and confirming my input.

The keyboard tab sequence is an innovative way to completely replace the mouse by allowing sight-impared users to follow a path across the screen with the help of the screen readers. It's incredible that screen reader and tab key technology enable user to completely

understand and fill out multiple online forms across several different screens without having to click on a single element. I think that screen readers and the tab key sequences were a breakthrough in assistive technology when they were first introduced, and I now more fully understand the importance of ensuring that your application meets the WCAG 2.1 AA compliance.